# WEN-HORNG SHEU

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## Education

**Ph.D. Candidate in Computer Science**, University of California, Davis; GPA: 4.0/4.0 *2023-2028 (expected)*

· Research Areas: Distributed Algorithms, Streaming Algorithms, Large-Scale Computation.

**M.S. in Computer Science**, National Tsing Hua University; GPA: 3.9/4.0 *2019-2021*

· Coursework: Approximation Algorithms, Computational Geometry.

**B.S. in Computer Science**, National Tsing Hua University; GPA: 3.85/4.0 *2015-2019*

· Coursework: Parallel Algorithm Design, Advanced Data Structures, Introduction to Machine Learning.

## Peer-Reviewed Publications

**Fixed-Threshold Peeling in Sublinear MPC: Round-Approximation Tradeoffs and Applications** DISC '26

· Designed distributed algorithms for the densest subgraph problem, with applications in community detection.

· Reduced round complexity in the Massively Parallel Computation (MPC) model via a graph exploration technique.

**Faster Semi-streaming Matchings via Alternating Trees** ICALP '25

· Designed a streaming algorithm for computing a $(1+\epsilon)$-approximate matching in memory-constrained settings.

· Improved a STOC '22 result, reducing the pass complexity from $O(\epsilon^{-19})$ to $O(\epsilon^{-6})$ via a new data structure.

**Faster MPC Algorithms for Approximate Allocation in Uniformly Sparse Graphs** SPAA '25

· In collaboration with a Google researcher, improved a distributed algorithm with applications in online advertising.

· Reduced round complexity from $O(\log n)$ to $\tilde{O}(\sqrt{\log n})$ using random sampling, extending an ICML '18 result.

**A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic** SPAA '25

· Proposed faster algorithms for computing a $(1+\epsilon)$-approximate matching in distributed and dynamic settings.

**A Kernelization Algorithm for Finding a Perfect Phylogeny from Mixed Tumor Samples** TCBB '25

· Developed a preprocessing procedure with applications in analyzing tumor samples; accelerated existing algorithms.

**Faster Algorithms for Constructing Frequency Difference Consensus Trees** TCBB '25

· Improved the time complexity of existing algorithms for identifying the common structure of phylogenetic trees.

**Parameterized Complexity for Finding a Perfect Phylogeny from Mixed Tumor Samples** SIDMA '23

· Proposed a faster algorithm for the split-row problem, with applications in analyzing tumor samples.

## Software Project

**Critical Path Reporting for Static Timing Analysis** (Second Place in ACM TAU 2018 Contest)

· Contributed to a C++ tool of 5,000+ lines of code for analyzing digital circuits specified in Verilog format.

· Implemented Eppstein's $k$-shortest path algorithm for enumerating critical paths through given logic gates.

## Selected Honors

**Shadow Program Committee Member** for ACM SPAA 2026

· Served on the committee for a parallel algorithms conference; reviewed 6 papers and participated in committee discussions.

**Gold Award** in the 2019 ICPC Asia Taipei-Hsinchu Regional Contest

· Competed in 6+ international programming contests, including multiple ICPC regional contests.

**Grandmaster** on Codeforces (max rating 2551)

· Ranked as a Grandmaster, within the top 1% globally on Codeforces, a competitive programming platform.

· Solved 1,500+ problems; placed in the top 1% among 10,000+ global contestants in 4 Codeforces contests.

## Skills

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| **Programming Languages** | C++, Python |
| **Tools** | Codex, Claude Code, Git, LaTeX, Linux/Bash, LiteLLM |
| **Languages** | English (fluent, TOEFL 105/120), Chinese (native) |
| **Coursera Certificate** | AI Agents and Agentic AI with Python & Generative AI |